

# Machine Learning for Customer Churn Prediction

Nelson

Javier

Nicolas Castro Rivera

## Abstract

Customer Churn is a problem about customer discontinue a company's services or products. Machine learning techniques can be applied with purpose of predict if a customer has probability of be churn and search reasons of why, this information can help the company to take decisions for not lost clients. This work search explore differents machine learning techniques to compare and get the best approach to solve this problem.

## 1 Introduction

Customer Churn refers to clients that potentially can discontinue a service or a product of a company. The main context for this project is the telecommunication

This should be a section of about one page. Here, the context of the problem should be fully described. Also, previous solutions to the same problem should be referenced. Most of the bibliography is cited here. While referencing previous solutions, you can also mention computational techniques used, which provides a complete reference for the reader. Usually, this section does not include images or tables.

## 2 Preprocessing data

Here you should describe the design of your solution, your technical decisions, and why you believe your solution is appropriate. Some general images are recommended to enhance your explanation, and this section is expected to be 1 to 2 pages long. It is not recommended to include code here, but if you are presenting a complex algorithm, you may add it. Remember to write full paragraphs, with one idea per paragraph, and avoid using itemized lists. Aim for clarity and readability. Include enough detail for reproducibility. Briefly mention any limitations or assumptions in your approach.

### 2.1 EDA

Dataset has the next features:

| Feature          | Description                                                                             |
|------------------|-----------------------------------------------------------------------------------------|
| customerID       | Unique value that identify a customer                                                   |
| gender           | Determines whether customer is Male or Female                                           |
| SeniorCitizen    | Determines whether a person is citizen or not                                           |
| Partner          | Determines whether person has partner                                                   |
| Dependents       | Determines whether customer has a dependent                                             |
| tenure           | Number of months has stayed with the company                                            |
| PhoneService     | Whether customer has phone service                                                      |
| MultipleLines    | Whether the customer has multiple lines, or one, or does not have phone service         |
| InternetService  | Internet service provider (Fiber optic, DSL and other)                                  |
| OnlineSecurity   | Whether customer has online security, or not, or does not have internet service         |
| OnlineBackup     | Whether customer has online backup, or not, or does not have internet service           |
| DeviceProtection | Whether customer has device protection, or not, or does not have internet service       |
| TechSupport      | Whether customer has tech support, or not, or does not have internet service            |
| StreamingTV      | Whether customer has streaming tv, or not, or does not have internet service            |
| StreamingMovies  | Whether customer has streaming movies, or not, or does not have internet service        |
| Contract         | Contract term duration (Month-to-month, One year, Two year)                             |
| PaperlessBilling | Whether customer has paperless billing or not                                           |
| PaymentMethod    | Payment method (Electronic check, Mailed check, Bank transfer (automatic), Credit card) |
| MonthlyCharges   | Value paid by customer monthly                                                          |
| TotalCharges     | Total value paid by customer                                                            |
| Churn            | Whether the customer churned or not, stopped using the service last month.              |

**Table 1:** Dataset features description

## 2.2 Subsection 2

Subsection example 1

## 2.3 Subsection 3

Subsection example 2

# 3 Results and Discussion

Explain the experiments you performed to demonstrate that your solution works. Summarize your unit tests in a paragraph, mentioning their philosophy, quantity, and results. Also, discuss any integration and acceptance tests to address software quality and user expectations. If possible, compare your results with other solutions using metrics. Use tables to summarize test definitions and results, and charts (such as box-plots or time series) for comparisons, depending on your project. Highlight the most important findings. Every figure and table must have a descriptive caption and be referenced in the text.

# 4 Conclusion

Write a couple of paragraphs to summarize your work, results, and achievements. Clearly state why your work was successful and how it helped to solve the problem. End with suggestions for future work or open questions that remain

# References

- [1] Rummery, G. A., & Niranjan, M. (1994). On-line Q-learning using connectionist systems. *Technical Report*, Cambridge University Engineering Department.